

CHURCH LAW FORD, United Kingdom

WMO: 035440

Lat: 52.3589N

Lon: 1.3314W

Elev: 106

StdP: 100.06

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.0	-2.6	-5.4	2.4	-1.2	-4.2	2.7	-0.5	11.4	8.0	10.2	7.8	1.6	30	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	26.8	18.7	24.6	17.6	22.8	16.7	19.5	24.5	18.4	22.9	17.5	21.5	3.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	12.8	21.0	16.7	12.0	20.0	15.8	11.4	19.1	55.9	24.9	52.3	23.0	49.5	21.6	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.0	7.8	6.9	-7.2	29.9	2.4	2.4	-9.0	31.7	-10.4	33.1	-11.7	34.5	-13.5	36.2
			-7.3	22.2	2.3	2.6	-9.0	24.1	-10.4	25.6	-11.7	27.1	-13.4	29.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	4.3	4.6	6.3	8.7	11.6	14.7	16.9	16.7	14.1	10.8	7.0	4.6
	DBStd	5.43	3.28	3.21	2.99	2.93	2.92	2.61	2.54	2.40	2.40	2.84	3.20	3.73
	HDD10.0	814	177	153	118	58	17	1	0	0	1	23	97	169
	HDD18.3	3054	435	386	373	290	209	114	61	63	128	232	339	425
	CDD10.0	836	1	1	4	18	66	141	213	208	124	49	8	3
	CDD18.3	34	0	0	0	0	0	4	16	13	2	0	0	0
	CDH23.3	322	0	0	0	1	6	50	153	97	14	2	0	0
	CDH26.7	58	0	0	0	0	0	7	31	19	1	0	0	0
(14) Wind	WSAvg	3.2	3.6	3.7	3.8	3.4	3.2	2.8	2.7	2.7	2.7	3.0	3.1	3.3
(15) Precipitation	PrecAvg	676	56	46	41	48	58	62	60	69	49	67	63	57
	PrecMax	859	140	76	80	117	105	143	132	140	120	114	100	105
	PrecMin	414	16	10	9	2	19	17	15	14	9	17	32	20
	PrecStd	118	30	19	19	32	23	35	30	31	29	28	20	24
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	13.4	16.3	21.2	24.1	27.6	29.7	29.1	25.4	20.0	15.2	13.5
		MCWB	10.8	11.2	11.0	14.0	17.0	19.2	19.7	20.4	17.9	16.0	13.5	11.9
	2%	DB	11.5	11.6	14.0	17.8	21.7	24.5	27.1	26.2	22.3	17.6	14.0	12.5
		MCWB	9.9	9.7	10.1	12.3	15.7	17.4	18.5	19.0	16.8	14.7	12.5	11.2
	5%	DB	10.4	10.4	12.5	15.7	19.5	22.3	24.8	23.6	20.3	16.3	12.9	11.3
		MCWB	8.9	8.7	9.4	11.2	14.5	16.4	17.5	17.3	15.5	13.9	11.4	10.1
	10%	DB	9.2	9.4	11.2	14.0	17.2	20.3	22.8	21.7	18.7	15.2	11.8	10.2
		MCWB	7.9	7.9	8.7	10.3	13.0	15.4	16.8	16.3	14.7	13.2	10.4	9.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	11.5	11.8	14.8	18.3	20.8	21.0	24.3	19.0	16.7	14.0	12.2
		MCDB	12.3	12.9	14.5	19.6	22.0	25.4	27.5	24.7	23.0	19.3	14.8	13.0
	2%	WB	10.2	10.1	10.9	13.1	16.4	18.3	19.4	19.3	17.5	15.3	12.8	11.4
		MCDB	11.3	11.3	13.1	16.6	20.5	22.4	24.9	24.3	20.9	17.0	13.8	12.3
	5%	WB	9.1	9.1	9.9	11.8	15.0	17.0	18.3	18.1	16.5	14.4	11.7	10.3
		MCDB	10.2	10.2	11.9	14.8	18.5	20.9	23.3	22.2	19.3	15.9	12.7	11.3
	10%	WB	8.1	8.0	9.0	10.7	13.8	16.0	17.4	17.1	15.4	13.4	10.6	9.1
		MCDB	9.1	9.2	11.0	13.6	16.7	19.6	21.8	20.5	17.9	14.9	11.6	10.1
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.7	7.2	8.7	9.0	9.2	9.8	9.2	8.4	6.7	5.5	5.0
		MCDBR	6.1	6.3	9.0	11.6	12.5	12.8	13.4	12.2	10.8	7.4	5.8	5.6
	5% WB	MCWBR	5.6	5.1	5.9	6.9	7.4	6.9	6.1	6.1	5.9	5.0	5.0	5.2
		MCWBR	5.9	6.2	8.0	10.5	11.0	11.1	11.6	10.5	9.3	6.8	5.7	5.5
(40) Clear-Sky Solar Irradiance	taub		0.322	0.337	0.370	0.396	0.401	0.400	0.398	0.386	0.368	0.350	0.324	0.311
		taud	2.356	2.381	2.287	2.225	2.250	2.279	2.289	2.349	2.400	2.430	2.407	2.336
	Ebn at Noon		667	774	813	835	850	853	847	839	811	748	661	608
		Edh at Noon	61	79	105	125	128	126	123	110	93	74	58	53
(44) All-Sky Solar Radiation	RadAvg		0.77	1.46	2.52	3.87	4.77	5.12	4.92	4.06	3.05	1.77	0.94	0.62
	RadStd		0.07	0.18	0.31	0.42	0.31	0.51	0.53	0.29	0.27	0.15	0.10	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (100 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page